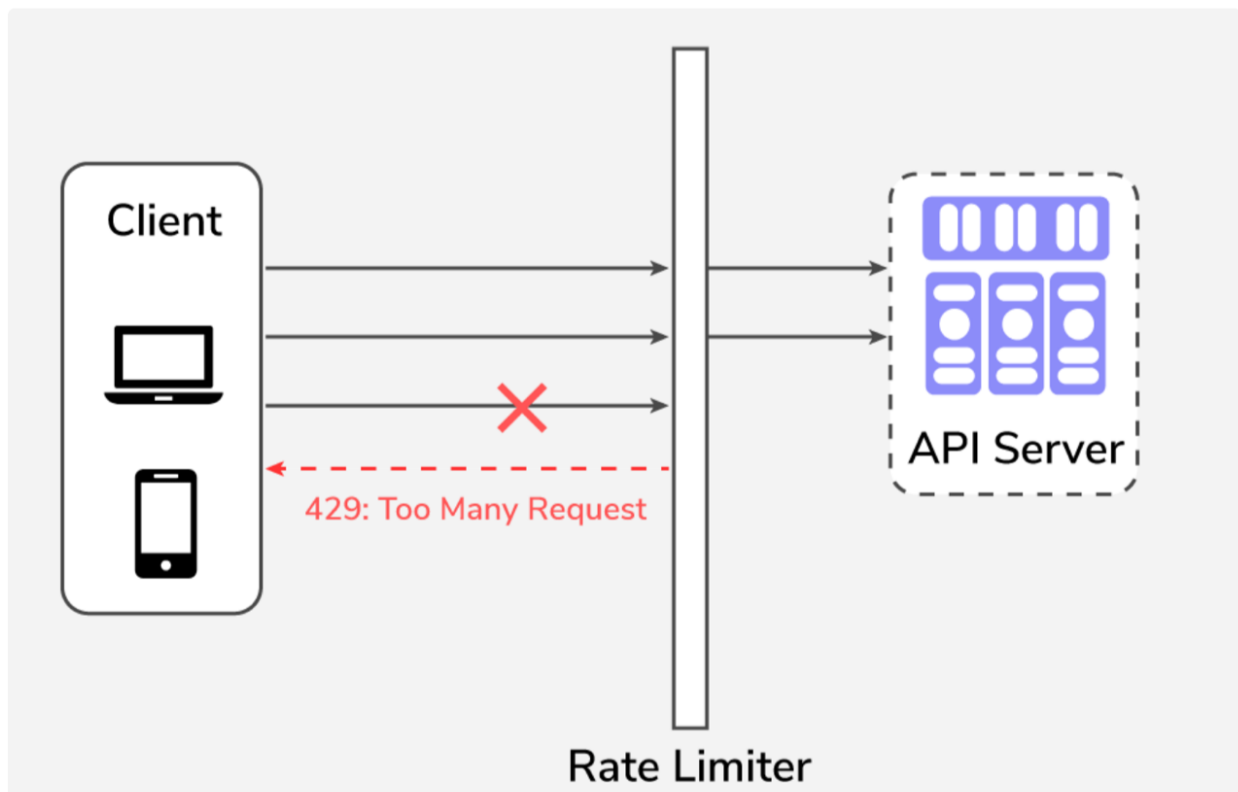


Rate Limiting is a technique used in system architecture to regulate how quickly a system processes or serves incoming requests or actions. It limits the quantity or frequency of client requests to prevent overload, maintain stability, and ensure fair resource distribution.

- Reduces the possibility of resource abuse and denial-of-service (DoS) attacks and hence the best possible performance, dependability, and security of systems.
- Frequently used in a variety of scenarios, including web servers, APIs, network traffic management, and database access.
- Examples of Rate Limiters are AWS API Gateway Throttling, AWS WAF (Web Application Firewall), Azure API Management Rate Limiting and Quota Policies and GCP's API Gateway Quota Policies.

A rate limiter is a component that controls the rate of traffic or requests to a system. It is a specific implementation or tool used to enforce rate-limiting.



Use Cases of Rate Limiting

Below are the use cases of Rate Limiting:

- **API Rate Limiting:** APIs commonly employ rate limitation to control the volume of client requests, ensure fair access to resources, and prevent abuse.
- **Web Server Rate Limiting:** Web servers employ rate limitation as a defense against denial-of-service attacks and to prevent server overload.
- **Database Rate limitation:** To keep the database server from experiencing undue strain and to preserve database performance, rate limitation is applied to database queries. For instance, to avoid resource exhaustion and guarantee seamless functioning, an e-commerce website can restrict the quantity of database queries per user.
- **Login Rate restriction:** To stop password guessing and brute-force assaults, login systems employ rate restriction. Systems can prevent unwanted access by restricting the quantity of login attempts made by each person or IP address.

Types of Rate Limiting

Below are the main types of rate limiting:

1. IP-based Rate Limiting

The number of requests that can be sent from a single IP address in a specified amount of time is limited by IP-based rate limitation (e.g., 10 requests per minute). By restricting the amount of traffic that can originate from a single IP address, it is frequently used to stop abuses like bots and denial-of-service attacks.

- **Advantages:**
 - It's simple to implement at both the network and application level.
 - If someone is trying to flood the system with requests, limiting their IP can prevent this.
- **Limitations:**
 - Attackers can use techniques like VPNs, proxy servers, or even botnets to spoof different IPs and get around the limit.
 - If multiple users share the same IP address (like in a corporate network), a legitimate user could get blocked if someone else on the same network exceeds the limit.

2. Server-based Rate Limiting

The number of requests that can be sent to a particular server in a predetermined period of time (e.g., 100 requests per second) is controlled by server-based rate restriction.

- **Advantages:**
 - Helps protect the server from being overwhelmed, especially during peak usage.
 - By limiting requests per server, you make sure that no single user can monopolize resources and degrade the experience for others.
- **Limitations:**
 - If attackers send requests across different servers, they might avoid hitting the rate limit on any single one.
 - If the limit is too low or the server is under heavy load, even legitimate users might face delays or blocks.

***For Example:** A music streaming service might implement server-based rate limiting to prevent their API from being overloaded during rush hours. By setting a limit of 100 requests per second per server, they can ensure the service stays fast and responsive even when usage spikes.*

3. Geography-based Rate Limiting

Geography-based rate limiting restricts traffic based on the geographic location of the IP address. It's useful for blocking malicious requests that originate from certain regions (for example, countries known for high levels of cyber attacks), or for complying with regional laws and regulations.

- **Advantages:**
 - If you know certain regions are the source of a lot of bad traffic (e.g., spam, hacking attempts), you can limit requests from those areas.
 - Helps comply with local data protection laws or restrictions on content in certain countries.
- **Limitations:**
 - Attackers can use VPNs or proxy servers to disguise their actual location and get around geography-based limits.
 - Users traveling or accessing services via international servers (like employees using VPNs or customers in hotels) might get blocked if they're in a region that's restricted.

***For Example:** A social media platform might implement geography-based rate limiting to fight spam. If a certain region is known for having a lot of fake accounts or bot activity, the platform could set a rule where IP addresses from that region can only make 10 requests per minute.*

